

Product Backlog

AttireAI - Team 11

Team:

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Problem Statement

Online clothing shoppers frequently struggle with selecting the right size and style, leading to high return rates and customer dissatisfaction. While existing fashion apps offer basic size charts or generic style suggestions, they fail to provide a personalized, end-to-end solution that considers individual body measurements, skin tone, hair color, and occasion-specific needs. Our web application AttireAI can address this gap by combining AI-powered body measurement technology with intelligent style recommendations, allowing users to receive personalized outfit suggestions complete with pricing and direct purchase links. Unlike competitors such as Stitch Fix or Amazon StyleSnap, AttireAI offers live measurement capture, comprehensive color analysis, and a virtual try-on feature that generates realistic outfit previews on the user's own image.

Background Information

a) Domain, Problem Background, and Target Users

E-commerce apparel shopping has a persistent “fit and style uncertainty” problem: users cannot reliably judge sizing and how an outfit will look on them using product photos alone. Body dimensions (e.g., chest, waist, hip, inseam), brand-to-brand sizing differences, and personal features such as skin tone and hair color strongly affect what fits well and what looks flattering. In addition, users shop for different social contexts (e.g., an interview, a formal dinner, a casual party), but most product pages do not translate those contexts into complete outfit choices.

Targeted users include:

- Online shoppers who frequently return clothing due to sizing issues or mismatched expectations.
- Fashion-conscious users who want guidance on what to wear for specific occasions.

- Users who want quick outfit bundles (top/bottom/shoes/accessories) instead of searching item-by-item.

b) Similar Existing Applications / Systems

Several systems partially address this space:

- Stitch Fix: provides styling assistance but typically depends on stylist-driven curation and scheduled shipping flows rather than live, interactive, user-controlled outfit generation.
- Amazon StyleSnap: supports image-based style search but is not designed as a measurement-driven sizing advisor or an occasion- and color-aware outfit builder.

c) Limitations of Other Solutions and How AttireAI Addresses Them

Limitation 1: Generic sizing help is not personalized enough.

- Many platforms rely on size charts and basic “recommended size” heuristics without capturing (or saving) detailed user measurements that can be reused across shopping sessions.
- AttireAI approach: Allow users to store detailed measurements and generate sizing guidance as part of the outfit recommendation workflow.

Limitation 2: Style suggestions are often not color-aware.

- Existing tools may suggest items based on trends or similarity, but do not explicitly incorporate skin tone and hair color, which strongly influence how colors appear on the user.
- AttireAI approach: Include optional skin tone and hair color analysis (and/or user selection) to prioritize flattering color palettes and reduce mismatched purchases.

Limitation 3: Outfit selection is not context-aware (occasion-specific).

- Users often need outfits for a specific event, but current systems frequently recommend individual items without building a complete outfit aligned to the formality level.
- AttireAI approach: Use an occasion-selection module (e.g., business meeting, date night, formal dinner) to generate complete outfit bundles appropriate for that setting.

Limitation 4: Try-on experiences are limited or not available for most products/users.

- Virtual try-on is not widely integrated end-to-end with personalized recommendations and is often separate from the shopping decision workflow.
- AttireAI approach: Provide a VIP tier that generates a realistic preview image for recommended outfits to improve confidence before purchase.

Requirements

Functional Requirements

User Authentication & Account Management

ID	User Story	Hours
1	As a user, I would like to create an account using my email address so that I can save my preferences and measurements.	6
2	As a user, I would like to log in using social login options (Google, Microsoft, Apple) for faster access to my account.	6
3	As a user, I would like to reset my password via email if I forget it.	2
4	As a user, I would like to log out of my account securely from any device.	1
5	As a user, I would like to view and edit my profile information (name, email, profile picture).	4
6	As a user, I would like to delete my account and all associated data if I no longer wish to use the service.	2

Body Measurements

ID	User Story	Hours
7	As a user, I would like to manually input my body measurements (waist, chest, hip, inseam, shoulder width, arm length, height, weight) so the system can recommend properly fitting clothes.	6
8	As a user, I would like to use my device's camera to automatically capture my body measurements through live detection technology.	36

9	As a user, I would like to receive on-screen guidance during the live measurement process to ensure accurate capture (e.g., "Stand 6 feet away", "Turn sideways").	8
10	As a user, I would like to review and confirm the automatically detected measurements before saving them.	2
11	As a user, I would like to edit individual measurements if the automatic detection was inaccurate.	3
12	As a user, I would like to save multiple measurement profiles (e.g., for different family members).	5
13	As a user, I would like the system to convert measurements between metric and imperial units based on my preference.	2
14	As a user, I would like to see which size I would be at different brands/retailers based on my measurements.	8

Color Analysis

ID	User Story	Hours
15	As a user, I would like to upload a photo of myself for skin tone analysis so that recommended outfits complement my complexion.	8
16	As a user, I would like to use my device's camera to capture my skin tone in real-time.	10
17	As a user, I would like the system to detect my hair color from an uploaded or captured photo.	6
18	As a user, I would like to manually select or adjust my skin tone category if the automatic detection is inaccurate.	2
19	As a user, I would like to manually select or adjust my hair color if the automatic detection is	2

	inaccurate.	
20	As a user, I would like to see my color profile summary (skin tone, hair color, recommended color palette) in my account.	4
21	As a user, I would like to receive recommendations for clothing colors that complement my skin tone and hair color.	6

Style Preferences

ID	User Story	Hours
22	As a user, I would like to select my preferred fashion styles from a list (minimalist, streetwear, classic, bohemian, preppy, athleisure, vintage, etc.) so recommendations match my taste.	4
23	As a user, I would like to select multiple style preferences to receive diverse outfit suggestions.	2
24	As a user, I would like to indicate styles I want to avoid so they are excluded from recommendations.	2
25	As a user, I would like to take a visual style quiz where I choose between outfit images to help the system understand my preferences.	12
26	As a user, I would like to update my style preferences at any time.	1
27	As a user, I would like to set preferred price ranges (budget, mid-range, luxury) for outfit recommendations.	3
28	As a user, I would like to specify preferred brands or retailers I like shopping from.	3
29	As a user, I would like to specify brands or retailers I want to exclude from recommendations.	2

Occasion Selection

ID	User Story	Hours
30	As a user, I would like to select an occasion type (formal dinner, casual party, business meeting, date night, wedding guest, job interview, workout, everyday casual, etc.) when requesting outfit recommendations.	4
31	As a user, I would like to specify the weather/season for the occasion so recommendations are weather-appropriate.	4
32	As a user, I would like to specify a dress code if applicable (black tie, business casual, smart casual, etc.).	2
33	As a user, I would like to save frequently used occasions as presets for quick access.	4
34	As a user, I would like to add custom occasion types if my specific need isn't listed.	2

AI Outfit Recommendations

ID	User Story	Hours
35	As a user, I would like to receive complete outfit recommendations based on my measurements, color profile, style preferences, and selected occasion.	25
36	As a user, I would like each outfit recommendation to include multiple clothing items that work together (top, bottom, shoes, accessories).	8
37	As a user, I would like to see multiple outfit options (at least 5) for each recommendation request so I can choose my favorite.	6
38	As a user, I would like to view the reasoning behind each recommendation (e.g., "This color complements your skin tone", "This cut flatters your body type").	8
39	As a user, I would like to rate outfit recommendations (like/dislike) so the system can	6

	learn my preferences over time.	
40	As a user, I would like to request new recommendations if I don't like the current suggestions.	2
41	As a user, I would like to filter recommendations by specific criteria (color, price, brand, item type).	6
42	As a user, I would like to see alternative items for any piece in a recommended outfit (e.g., "Show me other pants that would work with this outfit").	8
43	As a user, I would like the system to learn from my rating history and improve recommendations over time.	12

Product Information & Purchase

ID	User Story	Hours
44	As a user, I would like to see detailed product information for each recommended item (name, brand, material, care instructions).	4
45	As a user, I would like to see the price for each recommended item.	2
46	As a user, I would like to see which size I should order for each item based on my measurements.	6
47	As a user, I would like to click a direct purchase link that takes me to the retailer's website to buy the item.	4
48	As a user, I would like to see the total estimated cost of a complete outfit.	2
49	As a user, I would like to see if recommended items are in stock in my size.	6
50	As a user, I would like to be notified when items in my saved outfits go on sale.	8
51	As a user, I would like to compare prices across different retailers for the same or similar items.	8

Saved Items & History

ID	User Story	Hours
52	As a user, I would like to save outfit recommendations to a favorites/wishlist for future reference.	4
53	As a user, I would like to organize saved outfits into custom collections (e.g., "Work Outfits", "Weekend Looks", "Special Events").	6
54	As a user, I would like to view my recommendation history to revisit past suggestions.	4
55	As a user, I would like to remove items from my saved collections.	1
56	As a user, I would like to share saved outfits with friends via social media or messaging.	6
57	As a user, I would like to mark items I've already purchased to track my wardrobe.	4
58	As a user, I would like to receive outfit suggestions using items I've already purchased/marked as owned.	12

Virtual Try-On - VIP Feature (VIRTUAL TRY-ON)

ID	User Story	Hours
59	As a VIP user, I would like to upload a full-body photo of myself for virtual try-on purposes.	6
60	As a VIP user, I would like to see realistic preview images of recommended outfits rendered on my uploaded photo.	35
61	As a VIP user, I would like to try on individual items on my photo, not just complete outfits.	12
62	As a VIP user, I would like to save virtual try-on images to my account for later review.	4
63	As a VIP user, I would like to compare multiple	6

	virtual try-on images side by side.	
64	As a VIP user, I would like to share my virtual try-on images with friends for feedback.	4
65	As a VIP user, I would like to view virtual try-on results from different angles if multiple photos are provided.	15

VIP Subscription

ID	User Story	Hours
66	As a user, I would like to view the benefits of VIP subscription before subscribing.	2
67	As a user, I would like to subscribe to VIP membership to access virtual try-on and premium features.	8
68	As a VIP user, I would like to manage my subscription (view status, cancel, update payment method).	6
69	As a VIP user, I would like to receive unlimited outfit recommendations (vs. limited for free users).	4
70	As a user, I would like to receive a free trial of VIP features before committing to a subscription.	6

User Interface & Navigation (UI/UX)

ID	User Story	Hours
71	As a user, I would like a clean, intuitive homepage that guides me through the recommendation process.	8
72	As a user, I would like to access all main features from a clear navigation menu.	4
73	As a user, I would like the website to be responsive and work well on both desktop and mobile browsers.	12

74	As a user, I would like to see loading indicators when the system is processing my requests.	2
75	As a user, I would like to receive clear error messages if something goes wrong.	3
76	As a user, I would like to access a help/FAQ section to learn how to use the application.	4
77	As a user, I would like to complete an onboarding tutorial when I first create an account.	6
78	As a user, I would like to switch between light and dark mode themes.	4

Admin Features

ID	User Story	Hours
79	As an administrator, I would like to view user statistics and analytics (total users, active users, popular features).	8
80	As an administrator, I would like to manage the product database (add, edit, remove clothing items).	10
81	As an administrator, I would like to manage retailer partnerships and affiliate links.	6
82	As an administrator, I would like to review and respond to user feedback/support requests.	4

Non-Functional Requirements

Usability

- The application will have an intuitive, user-friendly interface with clear navigation across all main features (measurements, color analysis, outfit recommendations, virtual try-on).
- The application will be fully responsive and optimized for various device sizes (desktop, tablet, mobile) to ensure a consistent experience across platforms.
- The application will display helpful error messages and real-time guidance during critical workflows, particularly during live measurement capture and photo uploads.

- The application will provide clear visual feedback through loading indicators and progress bars during AI processing tasks.

Performance

- The homepage and primary navigation pages will load in under 3 seconds to maintain user engagement.
- AI outfit recommendations will be generated and displayed within 10 seconds of user request submission.
- Body measurement detection via camera will process and return results within 15 seconds to ensure a smooth user experience.
- Virtual try-on image generation will complete within 20 seconds to balance quality with acceptable wait times.
- The application will implement efficient caching strategies to reduce load times for frequently accessed data, such as saved outfits and user profiles.
- Image uploads will be optimized and compressed to ensure fast processing without sacrificing quality.

Scalability

- The application will be deployed on AWS cloud infrastructure that supports automatic scaling based on user demand and traffic patterns.
- The database architecture will be designed to handle growth from hundreds of concurrent users without performance degradation.
- The FastAPI/Flask backend will support horizontal scaling, allowing multiple instances to run simultaneously to handle increased load.
- The system will implement efficient data storage and retrieval practices to manage growing volumes of user profiles, measurement data, and outfit history.
- The application architecture will be modular to allow independent scaling of resource-intensive components like AI processing and image generation.

Security

- All communication between client and server will be encrypted using HTTPS/TLS to protect data in transit.
- User passwords will be hashed using bcrypt or similar industry-standard algorithms before storage, never stored in plain text.
- The application will implement secure session management with JWT tokens that expire after a period of inactivity.

- Access control will be enforced at the API level, ensuring users can only access and modify their own data through proper authentication and authorization checks.
- Uploaded photos and sensitive user data will be stored in secure AWS S3 buckets with appropriate access policies and encryption at rest.
- The application will implement input validation and sanitization on all user inputs to prevent SQL injection, XSS, and other common vulnerabilities.
- Payment processing for VIP subscriptions will be handled through PCI-compliant third-party payment processors (e.g., Stripe) to ensure no sensitive payment data is stored on our servers.
- User account deletion will permanently remove all associated personal data, including measurements, photos, and preferences, from the system.
- The FastAPI/Flask backend will implement rate limiting to prevent brute force attacks and API abuse.